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(54) **AIRBAG ASSEMBLIES FOR VEHICLES WITH GENEROUS LEG ROOM**

AIRBAGANORDNUNGEN FÜR FAHRZEUGE MIT GROSSZÜGIGEM BEINRAUM

ENSEMBLES COUSSIN DE SÉCURITÉ GONFLABLE POUR VÉHICULES AVEC LARGE ESPACE
POUR LES JAMBES

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Description

TECHNICAL FIELD

[0001] The present disclosure relates generally to the field of protection systems for vehicle occupants. More particularly, the present disclosure relates to airbag assemblies and other protection systems that are configured to cushion the knee and/or lower leg of an occupant during a collision event. Some airbag assemblies disclosed herein are suitable for use in vehicles that provide generous leg room to an occupant.

BACKGROUND

[0002] Protection systems have been installed in vehicles to protect an occupant during a collision event. Some protection systems suffer from one or more drawbacks or may perform less than optimally in one or more respects. Certain embodiments disclosed herein can address one or more of these issues.

[0003] Generic US 5,577,765 A shows an airbag assembly with a frontal airbag cushion and a knee airbag cushion according to the preamble of claim 1. In one embodiment, the frontal airbag cushion is only supported by the knee airbag cushion.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The written disclosure herein describes illustrative embodiments that are nonlimiting and non-exhaustive. Reference is made to certain of such illustrative embodiments that are depicted in the figures, in which:

FIG. 1 is a side view of an airbag assembly, according to one embodiment not belonging to the invention, prior to deployment.

FIG. 2 is a side view of the airbag assembly of FIG. 1 in a deployed state.

FIG. 3 is a perspective view of the airbag assembly of FIG. 1 in a deployed state.

FIG. 4 is a side view of an airbag assembly showing a tether, according to the invention.

DETAILED DESCRIPTION

[0005] The components of the embodiments as generally described and illustrated in the figures herein can be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the figures, is not intended to limit the scope of the present disclosure, but is merely representative of various embodiments. While various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0006] The phrase "coupled to" is used in its ordinary sense, and is broad enough to refer to any suitable coupling or other form of interaction between two or more entities, including mechanical, fluid, and thermal interaction. Two components may be coupled to each other even though they are not in direct contact with each other. The phrase "attached to" refers to interaction between two or more entities which are in direct contact with each other and/or are separated from each other only by a fastener of any suitable variety (e.g., mounting hardware or an adhesive). For the sake of clarity, an airbag cushion that shares a portion of a panel with another airbag cushion is "attached" to that airbag cushion.

[0007] As used herein, the terms "forward" and "rearward" are used with reference to the front and back of the relevant vehicle. For example, an airbag cushion that deploys in a rearward direction deploys toward the back of the vehicle. The term "frontal airbag," as used herein, refers to an airbag cushion that is configured to cushion the head and/or torso of an occupant as the occupant moves in a primarily forward direction during a collision event, regardless of whether the occupant is seated in a front seat of the vehicle.

[0008] As used herein, an "upper" or "lower" surface of an airbag cushion refers to an upper or lower surface of the airbag cushion when the airbag cushion is in an inflated state. The term "reaction surface," as used herein, refers to a surface disposed generally forward of an inflated airbag cushion that interacts with the airbag cushion to restrict the forward movement and/or forward rotation of an airbag cushion upon occupant loading. The term "occupant" refers to a person or crash test dummy within a vehicle. The term "knee restraint" refers to knee airbags and knee bolsters that are configured to cushion one or more knees of an occupant during a collision event. As used herein, the term "vehicle occupant position" refers to a position typically occupied by an occupant when seated in a seat of a vehicle. A vehicle occupant position may also be referred to as a "seating position," which refers to a position in which the vehicle is designed to have an occupant be positioned (e.g., seated) during operation.

[0009] Occupant protection systems may be installed at various locations within a vehicle to reduce or minimize occupant injury during a collision event. For example, such systems may be installed to cushion a front-seat passenger, a driver, or an occupant seated behind the front seat(s) of a vehicle. In the following disclosure, specific reference is made to occupant protection systems (e.g., airbag assemblies) that are designed to cushion a front-seat passenger, although the principles discussed herein may apply to automotive protective systems that are designed to cushion other occupants as well, such as the driver or occupants seated rearward of the front seat(s) of a vehicle. For instance, occupant protection systems may be installed in seats or other vehicle structures that are disposed forward of a back-seat occupant and operate in a manner analogous to that described in greater detail below in connection with protection systems for a front-seat passenger.

[0010] Some of the occupant protection systems disclosed herein may be particularly useful for protecting occupants that are seated in seats in a passenger compartment providing generous leg room. For example, in some embodiments disclosed herein, the distance from the instrument panel and the seat in which the occupant is seated is greater than or equal to 300, 350, 400, 500, 600, and/or 800 mm. In some embodiments, the distance between a housing disposed forward of the occupant and the seat in which the occupant is seated is greater than or equal to 350, 400, 500, 600 and/or 800 mm. The protection of occupants in such vehicles can present unique challenges due to the distance between the occupant's knees and the vehicle structure disposed directly forward of the occupant's knees. For occupants seated in a relatively cramped environment, a collision event often causes an occupant's knees to contact a fixed lower portion of the instrument panel which constitutes a knee bolster. This knee bolster can absorb impact energy imposed by the occupant's knees during a collision event. However, in vehicles that provide generous leg room, the instrument panel may be too far away from the occupant to absorb an occupant's impact energy.

[0011] Occupant protection systems often include one or more airbag cushions. Airbag cushions are typically disposed within a housing in a packaged state (e.g., rolled, folded, and/or otherwise compressed) and may be retained in the packaged state behind a cover. During a collision event, an inflator may be triggered, which rapidly fills an airbag cushion with inflation gas. The inflation gas may cause the airbag cushion to rapidly transition from a compact packaged (i.e., undeployed) state to an expanded or deployed state. In some embodiments, the expanding airbag cushion opens an airbag cover (e.g., by tearing through a tear seam or opening a door-like structure) to exit the housing. The inflator may be triggered by any suitable device or system, and the triggering may be in response to and/or influenced by one or more vehicle sensors.

[0012] Some occupant protection systems include a frontal airbag cushion that is configured to receive the head and/or torso of an occupant during a collision event. Some protection systems also include a knee restraint to protect the knees and/or lower leg of an occupant during a collision event. Such knee restraints may absorb the impact energy of an occupant, particularly during a frontal impact event. In some embodiments, a knee restraint includes a knee airbag cushion that contacts and cushions the knee during a collision event. In some embodiments, a knee restraint includes a knee bolster that absorbs the impact energy of an occupant during a collision event. Some knee bolsters disclosed herein are active knee bolsters that are capable of extending toward an occupant in anticipation of or in response to a collision event.

[0013] FIGS. 1-3 depict an airbag assembly 100. More specifically, FIG. 1 illustrates a side view of the airbag assembly 100 within a vehicle 10 prior to deployment,

while FIGS. 2 and 3 provide a side view and perspective view of the airbag assembly 100 in a deployed state.

[0014] As depicted in FIG. 1, the airbag assembly 100 includes an airbag housing 102 that is disposed and/or mounted within an instrument panel 70 forward of a vehicle occupant 50. In the depicted embodiment, the airbag housing 102 is disposed forward of an occupant 50 that is seated in a vehicle occupant position, within which the vehicle occupant 50 is generally positioned while in a seat 60 within the vehicle 10. In the depicted embodiment, the occupant 50 is seated in the seat 60 positioned within a passenger compartment of the vehicle 10 so as to provide generous leg room for the occupant 50. For example, in some embodiments, the distance (di) from the instrument panel 70 and the seat 60 in which the occupant 50 is seated is greater than or equal to 300, 350, 400, 500, 600, and/or 800 mm. In some embodiments, the distance (ck) between the airbag housing 102 and the seat 60 in which the occupant 50 is seated is greater than or equal to 350, 400, 500, 600, and/or 800 mm.

[0015] As shown in FIGS. 2 and 3, the airbag assembly 100 includes, in addition to the airbag housing 102, a knee airbag cushion 110 as a knee restraint, and a frontal airbag cushion 130.

[0016] The knee airbag cushion 110 is configured to receive inflation gas from an inflator 101 a and deploy rearward from the airbag housing 102 toward an occupant 50 to cushion one or more knees of the occupant 50 during a collision event. In some embodiments, the knee airbag cushion 110 extends a distance of greater than about 200, 250, and/or 300 mm from the airbag housing 102 to a region at which the knee airbag cushion 110 is disposed furthest from the airbag housing 102 when fully deployed.

[0017] The frontal airbag cushion 130 is configured to receive inflation gas from an inflator 101 b and deploy rearward from the airbag housing 102 to cushion one or more of a torso and a head of an occupant 50 as the occupant 50 moves forward relative to the vehicle 10 during a collision event. As depicted in FIGS. 2 and 3, the frontal airbag cushion 130, when fully deployed, is disposed generally above the knee airbag cushion 110 and extends over the lap of the occupant 50. In some embodiments, the frontal airbag cushion 130, when fully deployed, is disposed in front of the occupant 50 and at a position lower than the head of a fiftieth percentile adult male crash test dummy when the crash test dummy is seated in the vehicle occupant position. Examples of fiftieth percentile male crash test dummies include the Hybrid III 50th Percentile Male Crash Test Dummy and the THOR Advanced Crash Test Dummy. In some embodiments, the frontal airbag cushion 130, when fully deployed, extends a distance of greater than about 1 m from the airbag housing 102 to a region at which the frontal airbag cushion 130 is disposed furthest from the airbag housing 102.

[0018] The frontal airbag cushion 130 and the knee

airbag cushion 1 10 may be sized and shaped in any suitable manner. In some embodiments, the frontal airbag cushion 130 has a greater inflated volume than the knee airbag cushion 1 10. In other embodiments, the knee airbag cushion 1 10 has the same or a greater inflated volume than the frontal airbag cushion 130. In some embodiments, such as that depicted in FIGS. 2 and 3, the frontal airbag cushion 130 includes a pillow-shaped portion 132 for receiving one or more of the head or torso of an occupant 50, and a neck portion 134 that connects the pillow-shaped portion 132 to the airbag housing 102. More particularly, as shown in the depicted embodiment, the neck portion 134 may extend in a generally upward direction adjacent to a substantially vertical panel of the knee airbag cushion 1 10 to connect the pillow-shaped portion 132 of the frontal airbag cushion 130 to the airbag housing 102.

[0019] In some embodiments the frontal airbag cushion 130 shares at least a portion of a panel with the knee airbag cushion 1 10 to divide the frontal airbag cushion 130 from the knee airbag cushion 1 10. Described another way, in some embodiments, the frontal airbag cushion 130 and the knee airbag cushion 1 10 may be considered, in essence, a single, two-chambered airbag where the upper chamber and the lower chamber are separated by and/or share in common one or more panels. In other or further embodiments, a lower surface of the frontal airbag cushion 130 is attached to an upper surface of the knee airbag cushion 1 10. Such attachment may be via any suitable means. For example, the lower surface of the frontal airbag cushion 130 and the upper surface of the knee airbag cushion 1 10 may be attached to one another via stitching, adhesives, welding, or any other attachment mechanism.

[0020] With the frontal airbag cushion 130 and the knee airbag cushion 110 coupled to one another in this manner, the frontal airbag cushion 130 may restrain forward movement of the occupant 50 relative to the vehicle 10 during a collision event without relying on the instrument panel 70 or a windshield 80 as a reaction surface for stabilizing the frontal airbag cushion 130. Stated differently, the knee airbag cushion 1 10 may impede forward rotation or movement of the frontal airbag cushion 130 upon occupant 50 loading due to the attachment of the frontal airbag cushion 130 to the knee airbag cushion 110.

[0021] The airbag assembly 100 may be deployed to cushion an occupant 50 in connection with a collision event. For instance, during or in anticipation of a collision event, one or more vehicle sensors may be actuated, which trigger one or more inflators 101 a, 101 b that cause deployment of the knee airbag cushion 1 10 and the frontal airbag cushion 130 from the airbag housing 102. More specifically, in the embodiment depicted in FIG. 1, the knee airbag cushion inflator 101 a and the frontal airbag cushion inflator 101 b fill the corresponding airbag cushions 110, 130 with inflation gas, causing the airbag cushions 110, 130 to expand and emerge from the housing

102. In other embodiments, the frontal airbag cushion and the knee airbag cushion are inflated by the same inflator. This inflator may be a single-stage inflator or a two-stage inflator. In some embodiments, the knee airbag cushion 1 10 and the frontal airbag cushion 130 deploy essentially simultaneously. In other embodiments, either the knee airbag cushion 110 or the frontal airbag cushion 130 deploy prior to the other.

[0022] The knee airbag cushion 110 may contact and cushion one or more knees of an occupant 50, thereby pinning the knees and/or lower leg of the occupant 50 adjacent to the seat 60. Such contact, together with the forces on the occupant 50 generated by the collision event, may cause the occupant 50 to double over onto the frontal airbag cushion 130 in collision events that cause the occupant 50 to move in a forward direction relative to the vehicle 10. The frontal airbag cushion 130 may thus restrain the forward movement of the occupant 50 relative to the vehicle 10. Additionally, the knee airbag cushion 1 10, due to its attachment to the frontal airbag cushion 130, may restrict the forward movement of the frontal airbag cushion 130 upon occupant 50 loading.

[0023] FIG. 4 provides a side view of a deployed airbag assembly 600, according to another embodiment. The airbag assembly 600 includes an airbag housing 602, a knee airbag cushion 610 as a knee restraint, a frontal airbag cushion 630, and one or more tethers 690.

[0024] The airbag housing 602 is disposed generally forward of the occupant 50 when the occupant 50 is seated in the vehicle occupant position. In anticipation of or response to a collision event, the knee airbag cushion 610 and the frontal airbag cushion 630 emerge from the airbag housing 602 to absorb impact energy from the occupant 50. More particularly, the knee airbag cushion 610 receives inflation gas from an inflator 601a and deploys rearward from the airbag housing 602 toward an occupant 50 to cushion one or more knees of the occupant 50 during a collision event. The frontal airbag cushion 630 also receives inflation gas from an inflator 601b and deploys rearward from the airbag housing 602 to cushion one or more of a torso and a head of an occupant 50 as the occupant 50 moves forward relative to the vehicle 10 during the collision event.

[0025] In the embodiment depicted in FIG. 8, the frontal airbag cushion 630 is not attached to the knee airbag cushion 610, thereby allowing the frontal airbag cushion 630 to move independent of the knee airbag cushion 610. In other embodiments according to the invention, the knee airbag cushion is attached to the frontal airbag cushion as described above in connection with the airbag assembly 100 (e.g., via a shared panel, stitching, adhesives, welding, etc.).

[0026] In the depicted embodiment, the airbag assembly 600 includes one or more tethers 690 that restrict forward rotation of the frontal airbag cushion 630 (e.g., rotation toward the windshield 80 of a vehicle 10) during occupant 50 loading. The one or more tethers 690 may be attached to any suitable structure of the vehicle 10 to

limit the forward rotation of the frontal airbag cushion 630. In the depicted embodiment, the tether 690 includes a first end that is attached to the frontal airbag cushion 630. The tether 690 also includes a second end that is attached to the dashboard 70 adjacent the floor of the vehicle 10. In other embodiments, tether 690 may extend through one or more airbag cushions, with the second end of the tether attached to an airbag housing. The tether 690 restricts the upward movement of the airbag cushion 630, thereby limiting forward rotation of the airbag cushion 630 upon occupant loading. More particularly, the tether 690 restricts upward rotational movement of the airbag cushion 630 about the housing as a rotational axis.

[0027] In other embodiments in which the knee airbag cushion is attached to the frontal airbag cushion, the tether may be attached to the knee airbag cushion instead of the frontal airbag cushion. Upon occupant loading, such a tether restricts the upward movement of the knee airbag cushion, which in turn restricts the upward movement of the frontal airbag cushion due to the attachment of the knee airbag cushion to the frontal airbag cushion.

[0028] Reference throughout this specification to "an embodiment" or "the embodiment" means that a particular feature, structure, or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

Claims

1. A vehicle comprising an instrument panel (70), a windshield (80) and an airbag assembly (100, 600), said airbag assembly (100, 600) comprising:

a housing (102, 602) mounted in the vehicle;
a knee airbag cushion (110, 610) to deploy from the housing to cushion one or more knees of an occupant during a collision event of the vehicle;
and

a frontal airbag cushion (130, 630) to cushion one or more of a torso and a head of the occupant during the collision event;

wherein the frontal airbag cushion (130, 630) is attached to the knee airbag cushion (130, 630) to restrict forward movement of the frontal airbag cushion (130, 630) during occupant loading, wherein the frontal airbag cushion (130, 630) is configured to restrain forward movement of the occupant relative to the vehicle during the collision without relying on the instrument panel (70) or the windshield (80) as a reaction surface,
characterized in that

the frontal airbag cushion (130, 630) deploys from the housing (102, 602) and the fully deployed frontal airbag cushion (130,

630) is disposed above the knee airbag cushion (110, 610),

and further **characterized in that**

a tether (690) is provided that is coupled between a component of the vehicle and the knee airbag cushion (610) to limit movement of the knee airbag cushion, thereby limiting movement of the frontal airbag cushion (630) relative to the vehicle upon occupant impact, or

a tether (690) is provided that is coupled to the both the frontal airbag cushion (630) and to a component of the vehicle to secure the frontal airbag cushion (630) relative to the vehicle, thereby restricting movement of the frontal airbag cushion (630) relative to the vehicle during occupant loading, wherein it is preferred that a first end of the tether (690) is attached to the frontal airbag cushion (630) and a second end of the tether (690) is attached to one or more of an instrument panel, a frame, and a floor of the vehicle.

2. The vehicle of claim 1, wherein the knee airbag cushion (110, 610) is configured to, upon deployment, contact the one or more knees of the occupant, thereby causing the occupant to double over onto the frontal airbag cushion (130, 630) if the occupant is moving in a forward direction relative to the vehicle due to the collision event.
3. The vehicle of claim 1, wherein the distance between the housing (102, 602) and a seat in which the occupant is seated is greater than or equal to 350 mm.
4. The vehicle of claim 1, wherein the frontal airbag cushion (130, 630), when fully deployed, is disposed in front of the occupant and at a position lower than a head of a fiftieth percentile adult male crash test dummy when the crash test dummy is seated in the vehicle occupant position.
5. The vehicle of claim 1, wherein the frontal airbag cushion (130, 630) has a greater inflated volume than the knee airbag cushion.
6. The vehicle of claim 1, wherein the frontal airbag cushion (130, 630) is configured to extend over the lap of the occupant when fully deployed.

Patentansprüche

1. Fahrzeug, umfassend ein Armaturenbrett (70), eine Windschutzscheibe (80) und eine Airbaganordnung (100, 600), wobei die Airbaganordnung (100, 600) Folgendes umfasst:

ein Gehäuse (102, 602), das in dem Fahrzeug

gelagert ist;

ein Knieairbagkissen (110, 610), um sich aus dem Gehäuse zu entfalten, um ein oder mehrere Knie eines Insassen während eines Kollisionsereignisses des Fahrzeugs abzufedern; und
 ein Frontalairbagkissen (130, 630), um einen Torso und/oder einen Kopf des Insassen während des Kollisionsereignisses abzufedern;
 wobei das Frontalairbagkissen (130, 630) an dem Knieairbagkissen (130, 630) angebracht ist, um eine Vorwärtsbewegung des Frontalairbagkissens (130, 630) während der Insassenbelastung zu beschränken,
 wobei das Frontalairbagkissen (130, 630) konfiguriert ist, um die Vorwärtsbewegung des Insassen relativ zu dem Fahrzeug während der Kollision zu hemmen, ohne sich auf das Armaturenbrett (70) oder die Windschutzscheibe (80) als Reaktionsoberfläche zu verlassen,

dadurch gekennzeichnet, dass

das Frontalairbagkissen (130, 630) sich aus dem Gehäuse (102, 602) entfaltet und das vollständig entfaltete Frontalairbagkissen (130, 630) oberhalb des Knieairbagkissens (110, 610) angeordnet ist,
 und ferner **dadurch gekennzeichnet, dass:**

ein Haltegurt (690) bereitgestellt ist, der zwischen einer Komponente des Fahrzeugs und dem Knieairbagkissen (610) gekoppelt ist, um die Bewegung des Knieairbagkissens zu begrenzen, wobei dadurch die Bewegung des Frontalairbagkissens (630) relativ zu dem Fahrzeug beim Insassenaufprall begrenzt wird, oder

ein Haltegurt (690) bereitgestellt ist, der sowohl mit dem Frontalairbagkissen (630) als auch mit einer Komponente des Fahrzeugs gekoppelt ist, um das Frontalairbagkissen (630) relativ zu dem Fahrzeug zu sichern, wobei dadurch die Bewegung des Frontalairbagkissens (630) relativ zu dem Fahrzeug während der Insassenbelastung beschränkt wird, wobei es bevorzugt ist, dass ein erstes Ende des Haltegurts (690) an dem Frontalairbagkissen (630) angebracht ist und ein zweites Ende des Haltegurts (690) an einem Armaturenbrett, einem Rahmen und/oder einem Boden des Fahrzeugs angebracht ist.

2. Fahrzeug nach Anspruch 1, wobei das Knieairbagkissen (110, 610) konfiguriert ist, um bei einer Entfaltung, das eine oder die mehreren Knie des Fahrzeuginsassen zu berühren, wobei dadurch veranlasst wird, dass sich der Insasse auf das Frontalairbagkissen (130, 630) krümmt, falls sich der Insasse aufgrund des Kollisionsereignisses in einer Vor-

wärtsrichtung relativ zu dem Fahrzeug bewegt.

3. Fahrzeug nach Anspruch 1, wobei der Abstand zwischen dem Gehäuse (102, 602) und einem Sitz, in dem der Insasse sitzt, wenigstens 350 mm beträgt.
4. Fahrzeug nach Anspruch 1, wobei das Frontalairbagkissen (130, 630), wenn es vollständig entfaltet ist, vor dem Insassen und in einer Position angeordnet ist, die niedriger ist als ein Kopf eines Crashtest-Dummys, der den Abmaßen von 50 % der erwachsenen Männern entspricht, wenn der Crashtest-Dummy in der Fahrzeuginsassenposition sitzt.
5. Fahrzeug nach Anspruch 1, wobei das Frontalairbagkissen (130, 630) ein größeres aufgeblasenes Volumen als das Knieairbagkissen aufweist.
6. Fahrzeug nach Anspruch 1, wobei das Frontalairbagkissen (130, 630) konfiguriert ist, um sich über den Schoß des Insassen zu erstrecken, wenn es vollständig entfaltet ist.

25 Revendications

1. Véhicule comprenant un tableau de bord (70), un pare-brise (80) et un ensemble coussin de sécurité gonflable (100, 600), ledit ensemble coussin de sécurité gonflable (100, 600) comprenant :

un boîtier (102, 602) monté dans le véhicule ;
 un coussin d'un coussin de sécurité gonflable pour les genoux (110, 610) destiné à se déployer à partir du boîtier pour apporter un amortissement à un ou plusieurs genou(x) d'un occupant pendant un incident de collision du véhicule ; et
 un coussin d'un coussin de sécurité gonflable frontal (130, 630) pour apporter un amortissement au torse et/ou à la tête de l'occupant pendant l'incident de collision ;

le coussin du coussin de sécurité gonflable frontal (130, 630) étant fixé au coussin du coussin de sécurité gonflable pour les genoux (130, 630) de manière à restreindre le mouvement vers l'avant du coussin du coussin de sécurité gonflable frontal (130, 630) pendant le chargement de l'occupant,

le coussin du coussin de sécurité gonflable frontal (130, 630) étant conçu de manière à restreindre le mouvement vers l'avant de l'occupant par rapport au véhicule pendant la collision sans dépendre du tableau de bord (70) ou du pare-brise (80) comme surface de réaction,

caractérisé en ce que

le coussin du coussin de sécurité gonflable frontal (130, 630) se déploie à partir du boîtier (102, 602) et

- en ce que** le coussin du coussin de sécurité gonflable frontal (130, 630) entièrement déployé est disposé au-dessus du coussin du coussin de sécurité gonflable pour les genoux (110, 610),
 et caractérisé en outre
 ce qu'une sangle (690) est pourvue qui est accouplée entre un composant du véhicule et le coussin du coussin de sécurité gonflable pour les genoux (610) de manière à limiter le mouvement du coussin du coussin de sécurité gonflable pour les genoux, limitant ainsi le mouvement du coussin du coussin de sécurité gonflable frontal (630) par rapport au véhicule lors de l'impact de l'occupant, ou
en ce qu'une sangle (690) est pourvue qui est accouplée à la fois au coussin du coussin de sécurité gonflable frontal (630) et à un composant du véhicule pour fixer le coussin du coussin de sécurité gonflable frontal (630) par rapport au véhicule, limitant ainsi le mouvement du coussin du coussin de sécurité gonflable frontal (630) par rapport au véhicule pendant le chargement de l'occupant, étant préférable qu'une première extrémité de la sangle (690) soit fixée au coussin du coussin de sécurité gonflable frontal (630) et qu'une seconde extrémité de la sangle (690) soit fixée à un tableau de bord, et/ou un châssis et/ou un plancher du véhicule.
2. Véhicule selon la revendication 1, dans lequel le coussin du coussin de sécurité gonflable pour les genoux (110, 610) est conçu pour, lors de son déploiement, entrer en contact avec le ou les genoux de l'occupant, ce qui amène l'occupant à se pencher en avant sur le coussin du coussin gonflable frontal (130, 630) si l'occupant se déplace vers l'avant par rapport au véhicule en raison de l'incident de collision.
3. Véhicule selon la revendication 1, dans lequel la distance entre le boîtier (102, 602) et un siège dans lequel l'occupant est assis est supérieure ou égale à 350 mm.
4. Véhicule selon la revendication 1, dans lequel le coussin du coussin de sécurité gonflable frontal (130, 630), lorsqu'il est complètement déployé, est disposé devant l'occupant et dans une position plus basse que la tête d'un mannequin d'essai de choc d'homme adulte du cinquantième centile lorsque le mannequin d'essai de choc est assis en position d'occupant du véhicule.
5. Véhicule selon la revendication 1, dans lequel le coussin du coussin de sécurité gonflable frontal (130, 630) a un volume gonflé supérieur au coussin du coussin de sécurité gonflable pour les genoux.
6. Véhicule selon la revendication 1, dans lequel le coussin du coussin de sécurité gonflable frontal (130, 630) est conçu pour s'étendre sur les genoux de l'occupant lorsqu'il est complètement déployé.

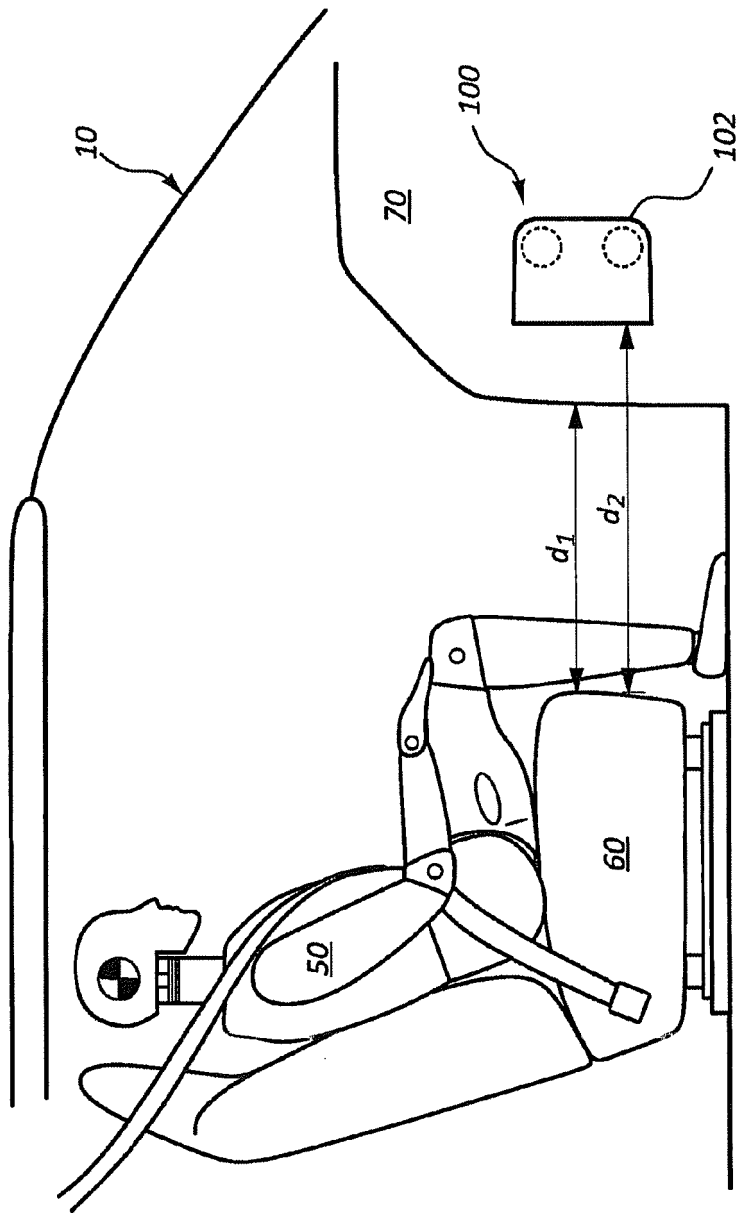


FIG. 1

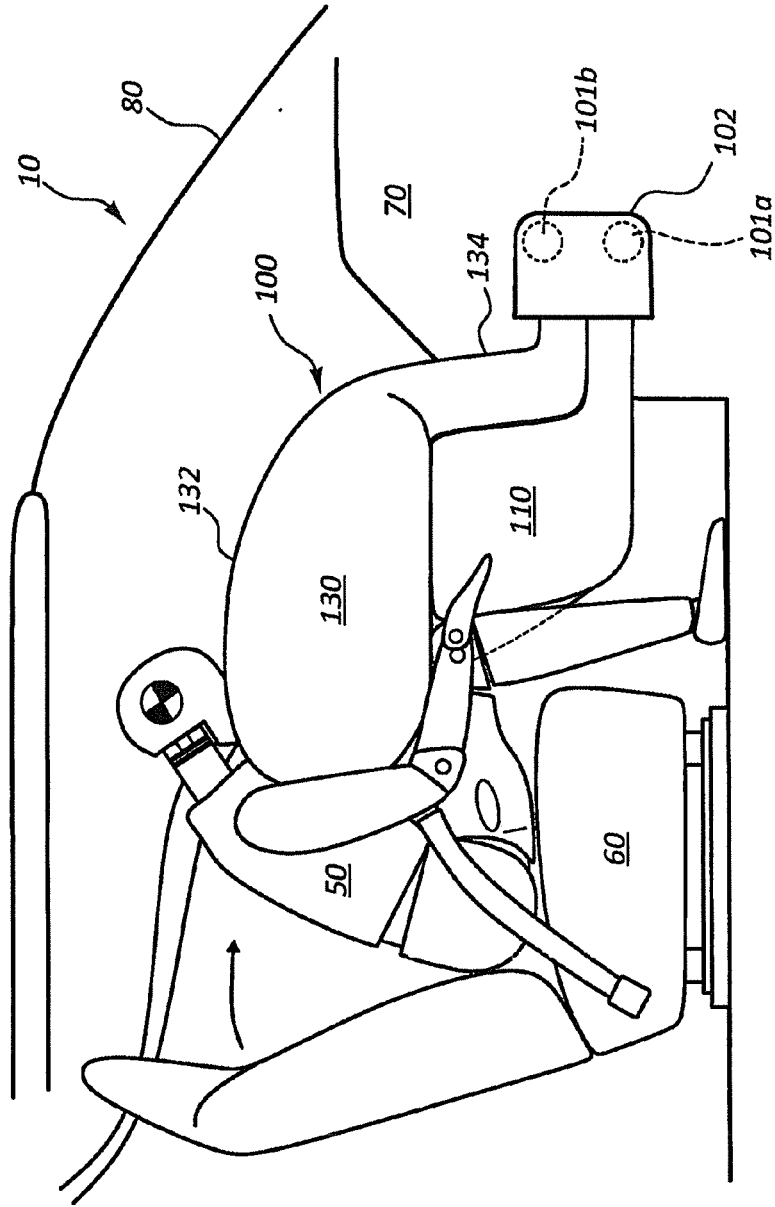


FIG. 2

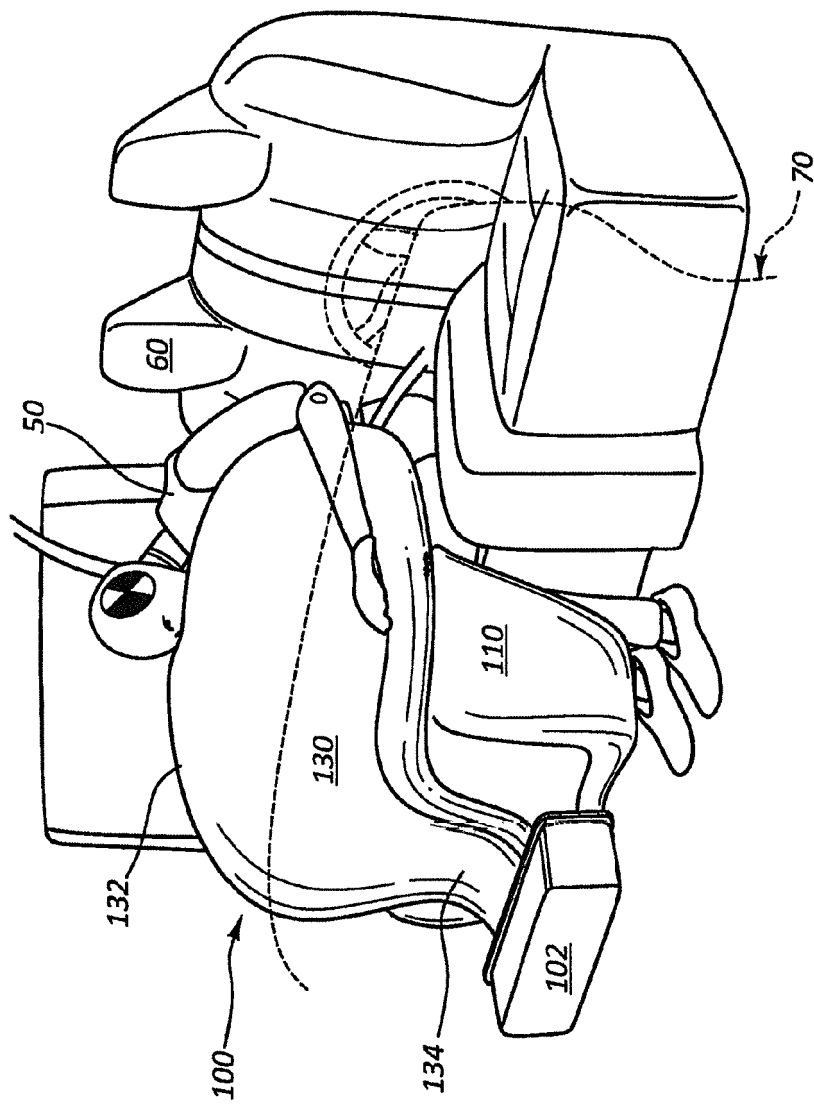


FIG. 3

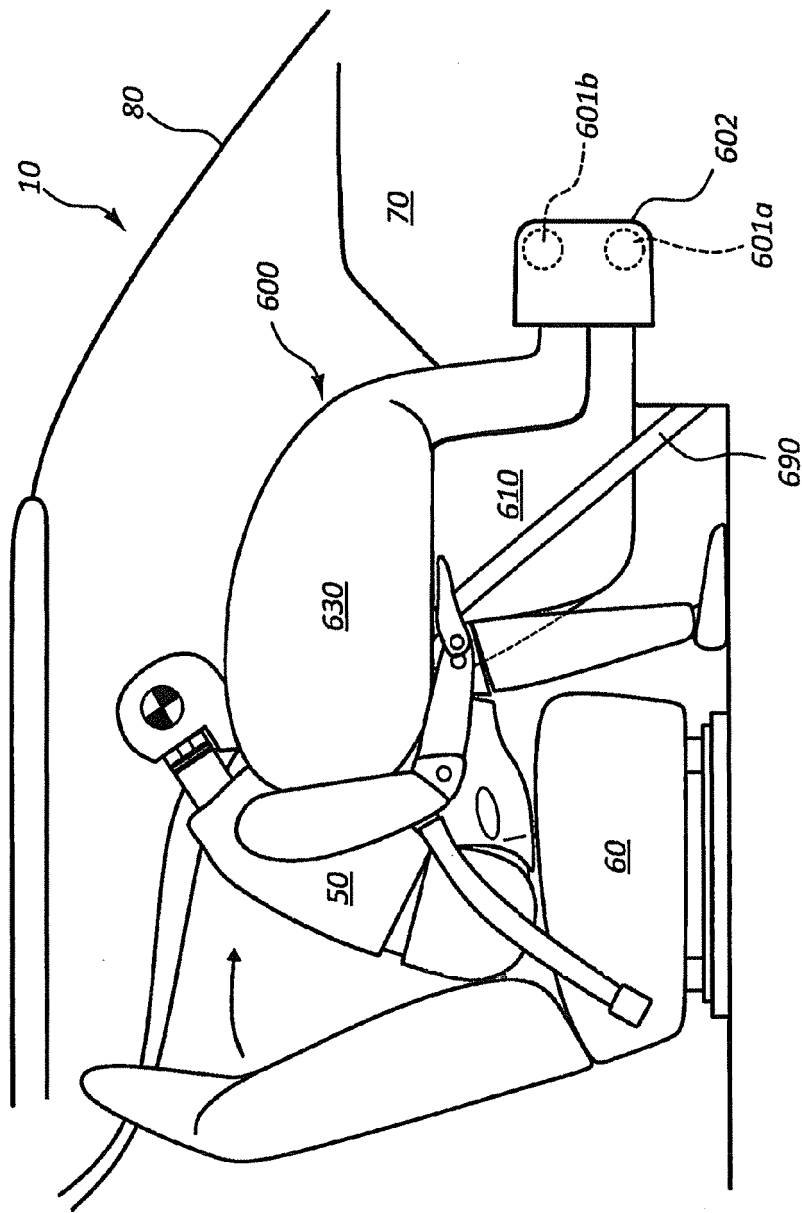


FIG. 4

REFERENCES CITED IN THE DESCRIPTION

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Patent documents cited in the description

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